

Divergence as the trace of a derivative

`MathlibAnnex.divergence`

def

Defines divergence without choosing coordinates.

Statement

Let E be a finite-dimensional real normed vector space and $W : E \rightarrow E$. The divergence of W at $x \in E$ is the trace of its real Fréchet derivative:

$$\operatorname{div} W(x) := \operatorname{tr}(DW(x)).$$

Definition

Here $DW(x)$ denotes the continuous linear derivative when it exists, and the zero map otherwise, following the source's total derivative convention. The RHS takes the trace of that endomorphism. Thus this definition still returns a value at a nondifferentiability point; it does not assert that the classical derivative exists there.

Assumptions

The definition accepts every function W and every point x . It imposes neither differentiability nor a measure. Finite dimensionality is part of the definition's domain; dimension zero is allowed.

Conclusion

The output is a real number. At a point where W is differentiable, it is the usual divergence. In coordinates $E = \mathbb{R}^\iota$ with a finite index set ι and coordinate vectors e_i ,

$$\operatorname{div} W(x) = \sum_{i \in \iota} (DW(x)e_i)_i.$$

For a differentiable field, writing $W_i(x) = (W(x))_i$ identifies the summand as $\partial_i W_i(x)$. The empty sum in dimension zero is zero.

The trace is independent of a basis. The coordinate identity follows by representing the derivative in the coordinate basis and summing its diagonal entries; the exact coordinate lemma is cited below. This identity also holds for the library's total derivative convention.

Main citations

- Exact declaration and its source — `MathlibAnnex.divergence`
- Trace as the coordinate divergence sum — `MathlibAnnex.divergence_pi`

Lean source signature (exact)

```
def divergence [FiniteDimensional ℝ E] (W : E → E) (x : E) : ℝ :=
  LinearMap.trace ℝ E (fderiv ℝ W x).toLinearMap
```

In the source	Mathematical meaning
<code>{E : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E]</code> <code>[FiniteDimensional ℝ E]</code>	The finite-dimensional real normed space E , without a chosen coordinate basis.
<code>(W : E → E) (x : E) : ℝ</code>	The arbitrary vector field W , evaluation point x , and real output $\operatorname{div} W(x)$.
In the source	Mathematical meaning
<code>fderiv ℝ W x</code>	The Fréchet derivative $DW(x)$ when it exists, and the zero continuous linear map otherwise. Differentiability is not a defining hypothesis.
<code>LinearMap.trace ℝ E (fderiv ℝ W x).toLinearMap</code>	The entire RHS $\operatorname{tr}(DW(x))$. The trace is basis independent; <code>.toLinearMap</code> forgets continuity while preserving the map.

▼ Exact surrounding binder context (separate excerpts)

Exact source lines 7–12:

```
noncomputable section
open Set
open scoped BigOperators

namespace MathlibAnnex
variable {E : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E]
```

▼ Exact content identity

Declaration: `MathlibAnnex.divergence`

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